

Table 10. Completed Data Charting Rows for the 68 Included Sources

Study ID	Source	Country or region	Evidence type	Main focus	Domain(s)	Integration level	Main charted finding
S001	Allison, 1954, Protection Afforded by Sickle-Cell Trait Against Subtertian Malarial Infection	East Africa	Foundational genetic epidemiology	Sickle cell trait and malaria	D1	Level 1	Provided early evidence that sickle cell trait confers protection against malaria in endemic African populations.
S002	Allison, 1954, The Distribution of the Sickle-Cell Trait in East Africa and Elsewhere	East Africa and broader African relevance	Foundational population genetics	HbS distribution and malaria ecology	D1, D2	Level 2	Linked the geographic distribution of sickle cell trait to malaria-endemic environments, supporting the malaria selection hypothesis.
S003	Aidoo et al., 2002, Protective Effects of the Sickle Cell Gene Against Malaria Morbidity and Mortality	Kenya	Cohort and case-control evidence	HbAS and malaria outcomes	D1	Level 1	Showed that the sickle cell gene was associated with protection against malaria morbidity and mortality among children in a malaria-endemic setting.
S004	Williams et al., 2005, Sickle Cell Trait and the Risk of <i>Plasmodium falciparum</i> Malaria and Other Childhood Diseases	Kenya	Cohort study	HbAS and <i>P. falciparum</i> malaria	D1	Level 1	Demonstrated that HbAS was associated with reduced risk of severe <i>P. falciparum</i> malaria and malaria-related childhood outcomes.
S005	Williams et al., 2005, Negative Epistasis Between HbAS and Alpha-Thalassaemia	Kenya	Genetic epidemiology	HbAS, alpha-thalassaemia, and epistasis	D1	Level 1	Reported that co-inheritance of HbAS and alpha-thalassaemia modified malaria protection, showing that single-variant models may be incomplete.
S006	Taylor et al., 2012, Haemoglobinopathies and the Clinical Epidemiology of Malaria	Global with SSA relevance	Review	Haemoglobin variants and malaria	D1	Level 1	Synthesised evidence that haemoglobin variants, including HbS and HbC, modify malaria susceptibility and severity.
S007	Taylor et al., 2013, Mechanisms of Malaria Protection by Sickle Cell Trait	Global with SSA relevance	Mechanistic review	HbAS malaria protection mechanisms	D1	Level 1	Summarised proposed mechanisms through which HbAS reduces severe malaria risk.
S008	Piel et al., 2010, Global Distribution of the Sickle Cell Gene and Geographical Confirmation of the Malaria Hypothesis	Global with strong Africa relevance	Bayesian geostatistical mapping	HbS distribution and malaria hypothesis	D1, D2	Level 2	Mapped HbS allele frequency and showed broad geographic correspondence with historical malaria endemicity.

S009	Piel et al., 2013, Global Epidemiology of Sickle Haemoglobin in Neonates	Global with strong SSA relevance	Geostatistical modelling	Neonatal HbS and HbSS births	D2	Level 1	Estimated neonatal HbS frequency and HbSS births, showing substantial spatial heterogeneity and high SCA birth burden in SSA.
S010	Piel et al., 2013, Global Burden of Sickle Cell Anaemia in Children Under Five	Global with strong SSA relevance	Burden modelling	SCA under-five burden	D2, D5	Level 2	Linked estimated SCA births to child survival and health-system need, especially in high-burden African settings.
S011	Modiano et al., 2001, Haemoglobin C Protects Against Clinical <i>P. falciparum</i> Malaria	Burkina Faso	Genetic epidemiology	HbC and clinical malaria	D1	Level 1	Showed that haemoglobin C was associated with protection against clinical <i>P. falciparum</i> malaria.
S012	Mockenhaupt et al., 2004, Hemoglobin C and Resistance to Severe Malaria in Ghanaian Children	Ghana	Case-control study	HbC, HbAS, and severe malaria	D1	Level 1	Reported associations between haemoglobin variants and resistance to severe malaria among Ghanaian children.
S013	Mockenhaupt et al., 2004, Alpha-Plus-Thalassaemia Protects African Children Against Severe Malaria	Ghana	Case-control study	Alpha-thalassaemia and severe malaria	D1	Level 1	Found evidence that alpha-plus-thalassaemia protected African children against severe malaria.
S014	Williams et al., 2005, Both Heterozygous and Homozygous Alpha-Plus-Thalassaemias Protect Against Severe and Fatal Malaria	Kenya	Case-control genetic epidemiology	Alpha-thalassaemia and severe malaria	D1	Level 1	Reported protection against severe and fatal malaria among children with alpha-plus-thalassaemia.
S015	Ruwende et al., 1995, Natural Selection of Hemi- and Heterozygotes for G6PD Deficiency by Resistance to Severe Malaria	Africa, multicountry relevance	Genetic epidemiology	G6PD deficiency and severe malaria	D1	Level 1	Provided evidence consistent with natural selection of G6PD deficiency through resistance to severe malaria.
S016	Mbanefo et al., 2017, Association of G6PD Deficiency and Malaria	Global with African relevance	Systematic review and meta-analysis	G6PD deficiency and malaria	D1	Level 1	Synthesised heterogeneous evidence on G6PD deficiency and malaria risk or severity.
S017	Rowe et al., 2007, Blood Group O Protects Against Severe <i>P. falciparum</i> Malaria Through Reduced Rosetting	Mali and Kenya	Genetic and mechanistic epidemiology	ABO blood group and rosetting	D1	Level 1	Showed that blood group O was associated with reduced rosetting and protection against severe malaria.
S018	Fry et al., 2008, ABO Glycosyltransferase Gene and Susceptibility to Severe Malaria	Gambia, Kenya, Malawi	Genetic association study	ABO variation and severe malaria	D1	Level 1	Reported association between ABO genetic variation and susceptibility to severe malaria

							across African populations.
S019	Kariuki et al., 2020, Red Blood Cell Tension Protects Against Severe Malaria in the Dantu Blood Group	Kenya	Genetic and mechanistic study	Dantu and parasite invasion	D1	Level 1	Showed that Dantu-associated red blood cell membrane tension reduced parasite invasion and protected against severe malaria.
S020	Malaria Genomic Epidemiology Network, 2014, Reappraisal of Known Malaria Resistance Loci	Multicountry Africa and Asia	Multicentre genetic association study	Malaria resistance loci	D1	Level 1	Reappraised known malaria resistance loci and confirmed the importance of host genetic variation in severe malaria risk.
S021	Weatherall, 2010, Genetic Variation and Susceptibility to Infection: The Red Cell and Malaria	Global with Africa relevance	Review	Red cell genetic variation and malaria	D1	Level 1	Provided broader synthesis of malaria-driven selection and red blood cell polymorphisms.
S022	Bhatt et al., 2015, The Effect of Malaria Control on <i>P. falciparum</i> in Africa Between 2000 and 2015	Endemic Africa	Geospatial modelling	Malaria burden and control	D2	Level 1	Showed that malaria burden declined over time with control interventions while remaining spatially heterogeneous.
S023	Gething et al., 2011, A New World Malaria Map: <i>P. falciparum</i> Endemicity in 2010	Global with strong Africa relevance	Geospatial modelling	Malaria endemicity	D2	Level 1	Mapped <i>P. falciparum</i> endemicity and demonstrated subnational variation in malaria exposure.
S024	Weiss et al., 2019, Mapping <i>P. falciparum</i> Mortality, Incidence, and Prevalence, 2000 to 2017	Global with strong SSA relevance	Geospatial modelling	Malaria prevalence, incidence, and mortality	D2	Level 1	Provided contemporary spatial estimates of malaria prevalence, incidence, and mortality, showing persistent geographic heterogeneity.
S025	Dalrymple et al., 2015, Malaria Mapping Using Routine Health Facility Data	Africa relevance	Spatial routine-data analysis	Routine malaria data and spatial estimation	D2, D4	Level 2	Showed that routine health facility malaria data can inform spatial burden estimation when interpreted with reporting and care-seeking processes.
S026	Alegana et al., 2016, Estimating Malaria Incidence Using Routine Health Facility Data	Kenya	Spatial routine-data modelling	Facility malaria data and catchments	D2, D4	Level 2	Estimated malaria incidence using routine facility data while accounting for treatment seeking and service catchment effects.
S027	Ndeezi et al., 2016, Uganda Sickle Surveillance Study	Uganda	National surveillance using dried blood spots	Sickle trait and SCD prevalence	D2, D4, D6	Level 4	Demonstrated that screening-linked surveillance can reveal subnational variation in sickle cell trait and SCD prevalence,

							creating evidence for spatial prioritisation and programme planning.
S028	Grosse et al., 2011, Sickle Cell Disease in Africa: A Neglected Cause of Early Childhood Mortality	Africa	Burden-focused review	SCD mortality and under-recognition	D4, D5	Level 2	Framed SCD as a neglected contributor to early childhood mortality in Africa and highlighted under-recognition before diagnosis or organised care.
S029	GBD 2021 Sickle Cell Disease Collaborators, 2023, Global, Regional, and National Prevalence and Mortality Burden of SCD	Global with strong SSA relevance	Burden and mortality attribution study	SCD prevalence and mortality	D4, D5	Level 3	Showed that total SCD-attributable mortality exceeds cause-specific SCD mortality because SCD contributes to deaths assigned to proximate causes.
S030	Kato et al., 2018, Sickle Cell Disease	Global	Review	SCD pathobiology and complications	D3, D5	Level 2	Summarised SCD pathophysiology, including haemolysis, vaso-occlusion, anaemia, and organ injury relevant to clinical vulnerability.
S031	White, 2018, Anaemia and Malaria	Global with malaria-endemic relevance	Review	Malaria-related anaemia	D3	Level 1	Summarised mechanisms through which malaria contributes to anaemia, supporting severe anaemia as a malaria-SCD convergence syndrome.
S032	Marsh et al., 1995, Indicators of Life-Threatening Malaria in African Children	Kenya	Clinical study	Severe malaria phenotypes	D3	Level 1	Identified clinical indicators of life-threatening malaria and supported phenotype-specific interpretation of severe malaria.
S033	Maitland et al., 2019, Immediate Transfusion in African Children With Uncomplicated Severe Anaemia	Uganda and Malawi	Randomised clinical trial	Severe anaemia and transfusion timing	D3, D5	Level 2	Established severe anaemia as an acute paediatric syndrome requiring transfusion-relevant service capacity.
S034	Maitland et al., 2019, Transfusion Volume for Children With Severe Anaemia in Africa	Uganda and Malawi	Randomised clinical trial	Severe anaemia and transfusion volume	D3, D5	Level 2	Showed that transfusion volume strategies are clinically important in the management of severe anaemia in African children.
S035	Uyoga et al., 2022, Sickle Cell Anaemia and Severe <i>P. falciparum</i> Malaria: Secondary Analysis of TRACT	Uganda	Secondary analysis of trial data with genotyping	SCA, severe anaemia, and malaria biomarkers	D3, D4	Level 4	Identified previously unknown SCA among children admitted with severe anaemia

							and showed genotype-specific differences in malaria biomarkers.
S036	Awotua-Efebo et al., 2004, Malaria Parasite Density and Sickle Cell Anaemia	Nigeria	Clinical comparative study	Malaria parasitaemia among children with SCA	D3	Level 1	Contributed clinical evidence on malaria parasite density among children with SCA.
S037	Nsagha et al., 2020, Malaria and Sickle Cell Disease Among Children in Cameroon	Cameroon	Hospital-based study	Malaria and SCD in children	D3	Level 1	Reported malaria prevalence, parasite density, and clinical patterns among children with SCD.
S038	Sumbele et al., 2016, Malaria and Moderate-to-Severe Anaemia in Cameroonian Children	Cameroon	Hospital-based study	Malaria and moderate-to-severe anaemia	D3	Level 1	Supported malaria-associated anaemia as a major paediatric clinical syndrome in an African hospital setting.
S039	Makani et al., 2011, Mortality in Sickle Cell Anaemia in Africa	Tanzania	Prospective cohort study	Mortality in SCA	D5	Level 1	Reported mortality among people with SCA in Tanzania, supporting the need for comprehensive care in African settings.
S040	Masamu et al., 2024, Very Severe Anaemia in Sickle Cell Disease in Tanzania	Tanzania	Cohort or hospital-based study	Very severe anaemia in SCD	D3, D5	Level 2	Showed that very severe anaemia is an important adverse clinical event among people with SCD in Tanzania.
S041	Gaston et al., 1986, Prophylaxis With Oral Penicillin in Children With Sickle Cell Anaemia	United States, foundational care evidence	Randomised clinical trial	Penicillin prophylaxis	D5	Level 1	Demonstrated that penicillin prophylaxis reduces pneumococcal septicemia in young children with SCA, supporting early diagnosis and preventive care.
S042	Quinn et al., 2010, Improved Survival of Children and Adolescents With Sickle Cell Disease	United States, comparative benchmark	Cohort study	Survival after organised SCD care	D5	Level 1	Provided benchmark evidence that newborn screening and organised care improve survival among children and adolescents with SCD.
S043	WHO, 2025, Sickle Cell Disease Fact Sheet	Global	Public health guidance	Essential SCD care	D5, D6	Level 2	Identified early diagnosis, vaccination, hydroxyurea, transfusion, and comprehensive management as key elements of SCD care.

S044	Tshilolo et al., 2019, Hydroxyurea for Children With Sickle Cell Anaemia in Sub-Saharan Africa	Angola, Democratic Republic of Congo, Kenya, Uganda	Open-label clinical trial	Hydroxyurea in children with SCA	D5, D6	Level 2	Showed that hydroxyurea was feasible, safe, and clinically beneficial for children with SCA in SSA.
S045	Opoka et al., 2017, Novel Use of Hydroxyurea in an African Region With Malaria	Uganda	Randomised placebo-controlled trial	Hydroxyurea in a malaria-endemic setting	D5, D6	Level 2	Addressed hydroxyurea safety and use among children with SCA in a malaria-endemic African setting.
S046	John et al., 2020, Hydroxyurea Dose Escalation for Sickle Cell Anaemia in Sub-Saharan Africa	Uganda	Randomised clinical trial	Hydroxyurea dose optimisation	D5, D6	Level 2	Added evidence on hydroxyurea dose escalation and optimisation among children with SCA in SSA.
S047	Tshilolo et al., 2009, Neonatal Screening for Sickle Cell Anaemia in the Democratic Republic of Congo	Democratic Republic of Congo	Newborn screening study	Neonatal SCA screening	D4, D6	Level 2	Demonstrated feasibility of neonatal screening for SCA in Central Africa.
S048	Rahimy et al., 2009, Newborn Screening for Sickle Cell Disease in the Republic of Benin	Benin	Newborn screening and follow-up programme	Screening, enrolment, and follow-up	D4, D5, D6	Level 3	Showed that newborn screening can be linked to enrolment and follow-up in an African programme setting.
S049	Odunvbun et al., 2008, Newborn Screening for Sickle Cell Disease in a Nigerian Hospital	Nigeria	Hospital-based newborn screening study	SCD screening feasibility	D4, D6	Level 2	Provided hospital-based evidence for newborn screening feasibility and SCD prevalence in Nigeria.
S050	Nnodu et al., Acceptability of Newborn Screening for Sickle Cell Disease in Nigeria	Nigeria	Multicentre survey	Acceptability of newborn screening	D6	Level 1	Reported acceptability of newborn screening for SCD, supporting feasibility of programme implementation.
S051	Ohene-Frempong et al., 2008, Screening Newborns for Sickle Cell Disease in Ghana	Ghana	Programme report	Newborn screening and tracking	D4, D6	Level 3	Described Ghana's newborn screening programme, including screening, tracking, result return, and clinic enrolment.
S052	Therrell et al., 2015, Newborn Screening for Sickle Cell Diseases in Africa	Africa	Review	Newborn screening implementation	D5, D6	Level 2	Synthesised African newborn screening experience and highlighted linkage-to-care and implementation barriers.
S053	McGann, 2014, Newborn Screening for Sickle Cell Disease in Africa	Africa	Review	Newborn screening priorities	D6	Level 1	Identified newborn screening as a public health priority for SCD in Africa.

S054	Green et al., 2022, The Consortium on Newborn Screening in Africa for Sickle Cell Disease	Multicountry SSA	Implementation research protocol	Screening-to-care implementation	D6	Level 4	Provided a multicountry implementation model linking newborn screening to standardised care and mortality ascertainment.
S055	Nnodu et al., 2024, Newborn Screening Initiatives for Sickle Cell Disease in Africa	Africa	Review	Screening initiatives, linkage, and barriers	D5, D6	Level 4	Synthesised African newborn screening initiatives and highlighted policy, process, enrolment, linkage, and sustainability gaps.
S056	Okeke et al., 2024, Sustainability of Newborn Screening for Sickle Cell Disease in Resource-Poor Countries	Resource-poor settings with Africa relevance	Systematic review	Screening sustainability	D6	Level 2	Identified policy, financing, workforce, laboratory, and follow-up barriers affecting newborn screening sustainability.
S057	Steele et al., 2019, Point-of-Care Screening for Sickle Cell Disease in Low-Resource Settings	Ghana, Martinique, United States	Diagnostic accuracy study	HemoTypeSC point-of-care testing	D6	Level 1	Showed that point-of-care testing can support SCD diagnosis in low-resource settings.
S058	Nnodu et al., 2019, HemoTypeSC Point-of-Care Testing: Promises and Challenges	Nigeria	Field evaluation and implementation study	Point-of-care SCD testing	D6	Level 2	Reported implementation experience with HemoTypeSC and highlighted practical challenges in decentralised SCD diagnosis.
S059	SickleInAfrica Consortium, 2020, SickleInAfrica: An African Research Network	Multicountry Africa	Consortium and network report	SCD registry and data infrastructure	D4, D6	Level 3	Described African SCD research infrastructure, registry development, and multicountry data-system strengthening.
S060	Makani et al., 2013, Sickle Cell Disease: New Opportunities and Challenges in Africa	Africa	Review	SCD burden, diagnosis, and care systems	D5, D6	Level 2	Identified opportunities and challenges for diagnosis, comprehensive care, and research in African SCD systems.
S061	Ware et al., 2017, Sickle Cell Disease	Global with SSA relevance	Review	SCD epidemiology and management	D5	Level 1	Summarised SCD burden, complications, and management approaches relevant to essential care planning.
S062	Wonkam et al., 2020, Sickle Cell Disease in Africa: Genomics and Public Health	Africa	Review or perspective	Genomics, diagnosis, and public health	D1, D5	Level 2	Linked African SCD burden to genomics, diagnosis, public health needs, and care gaps.

S063	Singer and Clair, 2003, Syndemics and Public Health	Global theoretical relevance	Theoretical paper	Syndemic theory	D7	Level 1	Defined syndemics as disease clustering and interaction within biosocial context.
S064	Singer et al., 2017, Syndemics and the Biosocial Conception of Health	Global theoretical relevance	Theoretical paper	Biosocial syndemic theory	D7	Level 1	Framed syndemics as interactions between diseases and harmful social or structural conditions.
S065	Tsai et al., 2017, Co-Occurring Epidemics, Syndemics, and HIV Risk	Global methodological relevance	Systematic review and methodological critique	Syndemic measurement	D7	Level 1	Highlighted limitations of co-occurrence-only syndemic claims and supported more careful evidence classification.
S066	Tsai, 2018, Syndemics: A Theory in Search of Data or Data in Search of a Theory?	Global methodological relevance	Methodological critique	Syndemic theory and evidence	D7	Level 1	Emphasised the need for analytic specificity when claiming syndemic interaction or excess burden.
S067	Mendenhall, 2017, Syndemics: A New Path for Global Health Research	Global theoretical relevance	Theoretical review	Syndemic theory in global health	D7	Level 1	Positioned syndemic theory as a global health framework for understanding interacting diseases and structural context.
S068	Mendenhall, 2021, Syndemics and Clinical Science	Global theoretical relevance	Conceptual review	Syndemic application and critique	D7	Level 1	Supported careful application of syndemic reasoning and cautioned against weak or underspecified syndemic claims.